

Detection and Classification of Small-sized UAVs and Birds in Sparse LiDAR Point Cloud*

Krisztián Balla[†], Anita Keszler[†], Sándor Gazdag[†] and András L. Majdik[†]

Abstract—Consumer drones have become significant safety hazards for critical infrastructure, disrupting airports and airplanes, while also raising privacy concerns, such as unauthorized recording. Hence, in this paper, we propose a cost-efficient method for detecting UAVs based on sparse point cloud data coming from a rosette scanning LiDAR sensor. Besides distinguishing between background and foreground elements, a major problem is filtering out false-positive detections due to birds and other flying objects. To address this issue, we integrate a classification network into our solution as well as a semantic segmentation network that is tasked with detecting potential threats. Our system utilizes the attention mechanism throughout the object detection phase to efficiently capture the global features of the scene, while it benefits from the convolutional layer’s local feature maps during classification. Experimental results show that our solution is capable of real-time object detection and recognition while maintaining its robustness.

I. MOTIVATION

Performance evaluation Speed evaluation Time taken

In order to mitigate the threats represented by the inappropriate use of commercially available small rotorcraft UAVs, this paper presents a method to identify these objects to protect critical infrastructure and privacy. The detection and classification of drones can be achieved on different modalities of data, including information from RADAR [1] and camera [2]–[4], which were previously in the focus in this research field. Unfortunately, the former typically demands military-grade hardware, while the latter is unable to deliver 3D information on the object’s position. Novel LiDAR sensors such as the LiVOX AVIA¹ offer a consumer-grade alternative capable of delivering spatial information. Unlike the more commonly used repetitive line-scanning LiDARs, the Avia employs a non-repetitive circular scanning pattern, which greatly enhances 3D coverage and range. However, due to this unique pattern, much of the resulting point cloud is sparse, making it both challenging and research-wise compelling to develop solutions capable of detecting and classifying objects based on this data.

II. RELATED WORKS

In this section, we provide a brief summary of recent research findings on detecting, tracking, and potentially classifying small-sized UAVs using point clouds.

In our previous work [5] a probabilistic 3D map (octomap [6], [7]) is used as the representation of the static background. This method requires a short initialization phase to build the

background model which is then subtracted from the subsequent incoming point clouds, leaving only the dynamic parts of it. The employed method of static background removal has its drawbacks. Dynamic changes in the background cause false detections, and the lack of classification of dynamic objects means the filter follows any moving object. Another similar method is [8] which also lacks the capability of differentiating UAVs from birds.

A custom-trained yolov5 [9] model is employed by [10] and [11] on the range image generated from the LiDAR measurements. These methods require a large number of points returned from the drone to function so they only work when the drone is close to the sensor.

In [12] flying (e.g. drones or birds) and stationary objects are distinguished by the identification of typical movement patterns in the trajectory of the tracked object. The authors track each possible object for a given amount of frames and if enough movement happens they label the object as a threat. However, identifying and classifying the flying objects remain an unsolved problem (e.g. a bird’s flight pattern fits their definition of threat and would be classified as one). To address this issue, [13] presents a method that leverages a grayscale camera on the pan-tilt system to classify the detected flying object with a Faster R-CNN up to a maximum distance of 40 m. In contrast, our research solves the detection and classification problem using only LiDAR modality on objects up to 100 m away.

III. METHODOLOGY

A. Overview

The general pipeline of our proposed solution is illustrated in Fig. 1. The raw point cloud is processed by a semantic segmentation network, which removes both static and dynamic background objects from the scene. The remaining foreground points are then clustered using K-Means and a Nearest Neighbour Search (NNS) is performed on the original point cloud based on each cluster’s mean coordinate value to maximize the segmentation accuracy. The radius selected for the NNS was determined by the average size of the objects in our dataset, which is approximately 1 to 1.5 meters in diameter. Finally, the clusters with sufficient number of points inside them are processed by a customized and re-trained PointNet [14], which assigns either the drone or bird labels to them.

B. Attention-based Semantic Segmentation

Fig. 2 shows the architecture of the designed model, which incorporates a spatial transformer [15] (T-Net) subnetwork,

[†]HUN-REN Institute for Computer Science and Control, 1111 Budapest, Hungary, firstname.lastname@szaki.hun-ren.hu

¹<https://www.livoxtech.com/avia>

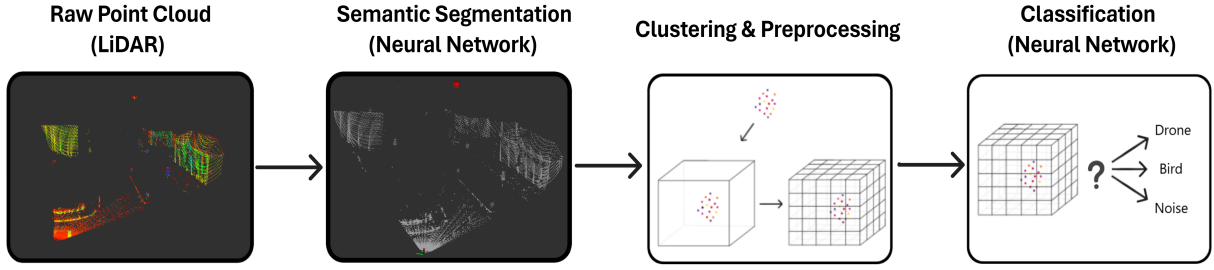


Fig. 1. Outline of the proposed solution: The raw point cloud stream, that is provided by a rosette scanning LiDAR serves as input for an attention-based neural network that removes both the static and dynamic background elements from it. The remaining foreground points then undergo a clustering and a unique preprocessing step in order to be classified by a slightly modified PointNet network.

the encoder part of a transformer, and a 1-Dimensional convolutional neural network (also referred to as Multi-Layer Perceptron - MLP). The central component of our model is the multi-head attention mechanism within the encoder layer, a choice strongly influenced by the recent uprising of the attention-based semantic segmentation networks. The ADE20K [16] is amongst the most prestigious challenges in this field, where the superiority of the transformer-encoder-based solutions is undeniable².

One of the key challenges when working with point clouds is the network’s susceptibility to rotations and translations present in the input cloud. Similarly to the solution in PointNet, we assume a point cloud $\mathbf{X} \in \mathbb{R}^{n \times 4}$, which consists of n points with $\{x, y, z, reflectivity\}$ fields. Since the reflectivity value does not carry spatial information, the point cloud is sliced into a reflectivity vector $\mathbf{X}_r \in \mathbb{R}^{n \times 1}$ and a position matrix $\mathbf{X}_p \in \mathbb{R}^{n \times 3}$. To compensate for any transformations that might be present in the latter matrix, we use a spatial transformer, which generates a transformation matrix $\mathbf{T} \in \mathbb{R}^{3 \times 3}$ (the parameters are learnable and adjusted through training), that projects the positions into a uniform space $\mathbf{X}_{p,T} \in \mathbb{R}^{n \times 3} = \mathbf{X}_p \mathbf{T}$, thus allowing for a more general representation, regardless of the initial transformation. Finally, we concatenate the corresponding reflectivity values to the transformed position matrix, resulting in a point cloud that is interpreted in a less sensitive domain:

$$\mathbf{X}_T \in \mathbb{R}^{n \times 4} = \mathbf{X}_p \cdot f(\mathbf{X}_p, \Psi) \parallel \mathbf{X}_r \quad (1)$$

where f is the spatial transformer, and Ψ represents its weights and biases.

In the new, transformation invariant 3D space, we extend our initial point features $\{x, y, z, reflectivity\}$ with learnable embeddings $\mathbf{E} \in \mathbb{R}^{n \times 28}$, resulting in tokens $\mathbf{X}_t \in \mathbb{R}^{n \times 32}$. Unlike other methods, that rasterize the 3D space and handle each voxel as a token, we skip the voxelization step and directly use the individual points as tokens. Although this approach is computationally more expensive due to the larger matrices in Multi-Head Attention, it achieves faster inference speeds. This is because the time saved by avoiding rasterization far outweighs the overhead introduced by the larger matrices in CUDA-supported tensor multiplications. From these tokens, the point embeddings

$\mathbf{X}_e \in \mathbb{R}^{n \times 32}$ are created with a linear projection:

$$\mathbf{P} : \mathbf{X}_t \mapsto \mathbf{X}_e \in \mathbb{R}^{n \times 32} \quad (2)$$

where $\mathbf{P}$ is the linear projection. These point embeddings serve as an input for the encoder, where the data is split into m different heads that capture relationships at different levels between the input and output tokens resulting in matrices $\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h \in \mathbb{R}^{n \times m \times (32/m)}$. In our case, the query q is a single point from the cloud (in practice, the attention function is calculated on a set of queries collected into matrix Q simultaneously), while the key K is the set of all points. From these two matrices the attention weight $\mathbf{w}_a \in \mathbb{R}^{n \times m \times m}$ is calculated [17], which helps to identify the set of tokens (points) that contribute the most to the classification of a given input token. Multiplying these weights by the values V and concatenating the different heads together results in the attention matrix $\mathbf{A} \in \mathbb{R}^{n \times 32}$. After layer normalization and a feed-forward network, our new data representation captures the global context of the tokens, which yields a robust, scene-wide awareness. Based on this global relationship map, the output class of either background or foreground is assigned to each point by a series of 1D convolutional layers.

C. Preprocessing & Classification

Exploiting the classes assigned to each point by the semantic segmentation network, the background points are removed from the cloud, while the foreground points undergo a clustering process. To further improve our solution’s robustness against false detections, the clusters with less than 3 points are discarded, as they just as likely to be sensor noise as actual objects. The mean of the coordinates is then calculated for each remaining cluster, based on which an NNS is executed on the original raw point cloud. This method greatly improves our ability to recover foreground points that were falsely classified as background ones in the segmentation process. Following this step, the often used approach (e.g. in [18]) is to use zero padding for upsampling and random sampling for downsampling, as the classification network needs a uniformly sized input. Our solution’s aim is to efficiently classify small aerial objects based on sparse point cloud data, which means that the number of reflections from a given object on certain occasions could approach zero. This led us to the conclusion, that we cannot allow to

²<https://paperswithcode.com/sota/semantic-segmentation-on-ade20k>

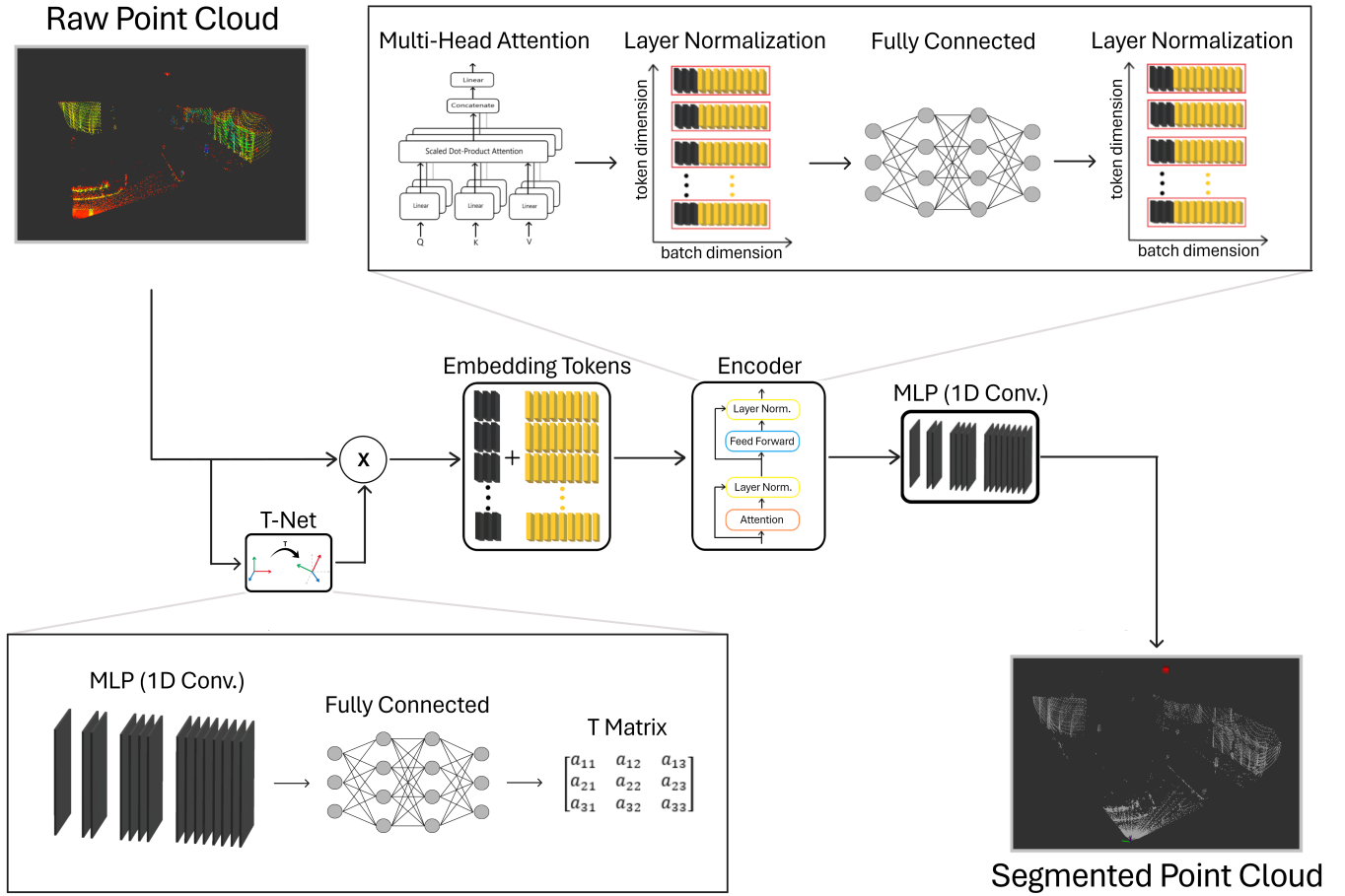


Fig. 2. Overview of the semantic segmentation network: the point cloud is transformed into a transformation invariant space, where each of the points influence is calculated with relevance to all the other points through the Multi-Head attention [17] mechanism and finally, all of them are assigned with a class determined by an MLP.

either deform the original shape or even worse, lose points that might carry crucial information. With this in mind, our proposed sampling method (see the right part of Fig. 1) encloses each cluster $C_i \in \mathbb{R}^{m_i \times 4}$ with a cube $Q_i \in \mathbb{R}^3$ in 3-Dimensions. We then further divide the enclosed space into smaller cubes, thus creating a voxel grid $V_i \in \mathbb{R}^{l \times w \times h}$ with uniform resolution in all dimensions ($l = w = h$). Afterward, we examine each voxel to determine if it contains any of the original points from the cluster. If it's empty, then the reflectivity value is set to zero, otherwise, it inherits the mean of the reflectivity across all of the points that it encloses. These voxel grids not only guarantee a uniform size, but a uniform 3D shape as well, which experimentally improves the generalization abilities of a slightly modified PointNet [14], that classifies them as either *drone*, *bird* or *unknown*.

IV. RESULTS AND EXPERIMENTS

An important requirement for our proposed single-frame detector is its ability to operate in real-time applications. The LiVOX Avia sensor in single-return mode generates a point cloud of 24000 points every 100 milliseconds, therefore our application should operate at speeds higher than 10 Frames

Per Seconds (FPS). Preliminary results on 1100 point clouds with 6 different scenes (both inside and outside) show processing speeds approaching 60 FPS, leaving sufficient room for future functionalities and algorithms. The performance of the individual components can be seen in Table I.

Furthermore in TABLE II, we benchmarked the speed our proposed dynamic background segmentator against a static algorithm presented in [5], where an octree-based background subtraction is followed by a statistical outlier and a radius outlier filter. Our method not only handles dynamic environments effectively but also demonstrates a significantly faster runtime with much lower deviation.

TABLE I
COMPONENT-WISE PERFORMANCE REGARDING INFERENCE SPEED

	Segmentation	Clustering	Preprocessing	Classification
Worst-case	6.96 ms	8.86 ms	16.82 ms	8.14 ms
Deviation	0.49 ms	1.41 ms	2.46 ms	0.39 ms
Mean	4.29 ms	1.65 ms	4.11 ms	6.44 ms

Due to the nature of the task and the point clouds, the semantic segmentation can be interpreted as an imbalanced classification problem, where one class (environment) over-

TABLE II
STATIC AND DYNAMIC BACKGROUND REMOVAL SPEED COMPARISON

	Dynamic (ours)	Static [5]
Worst-case	6.96 ms	57.27 ms
Deviation	0.49 ms	9.65 ms
Mean	4.29 ms	16.49 ms

powers the other (flying objects). Our chosen metric to accurately evaluate the model’s performance is the balanced F-Measure (also known as F_1), which is the harmonic mean of the precision and recall. Table III. displays the aforementioned performance metrics obtained from the test dataset. For our application, an F-Measure of ~ 0.78 proved to be satisfactory. The low number of false positives ensured that at least some parts of the objects were found, based on which we were able to recover the whole object by running an NNS on the original point cloud. Furthermore, false negative detections were easily filtered out since they primarily appeared as sensor noise scattered in small clusters of one to two points throughout the scene.

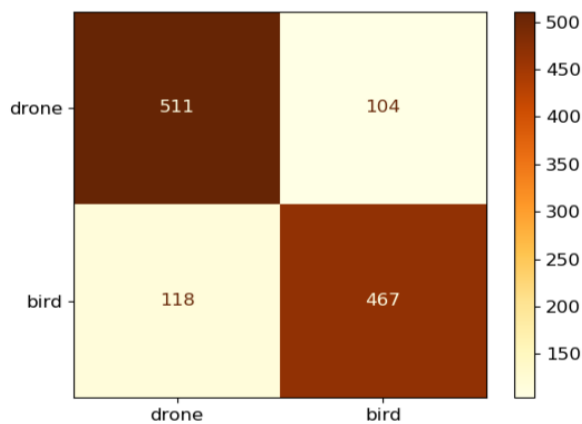


Fig. 3. Confusion matrix on the test dataset for the classification network.

TABLE III
PERFORMANCE EVALUATION OF THE SEMANTIC SEGMENTATION NETWORK

Precision	Recall	F-Measure
0.7634	0.7949	0.7787

Finally, the classification network was evaluated not as a standalone component, but as the last part of the entire solution. This implies that if any of the preceding components had errors (of any magnitude), the classification network had to operate with this imperfect data. As a result, the performance and output of the classification network were evaluated in the context of the overall solution. The confusion matrix generated from the tests can be seen in Fig. 3. Given that our dataset had a balanced distribution of drones and birds, accuracy is the most appropriate metric for evaluation, which averaged **81.5%**. We also assessed our

solution’s performance by comparing its effective range with the classification accuracy, as illustrated in Fig. 4.

A more thorough assessment of our detection range and accuracy will be completed by the final submission deadline. This evaluation will include a probabilistic analysis about determining the optimal minimum number of reflections required from an object to further process it (rather than labeling it as noise). It will also feature an in-depth examination of where and why our solution fails to either detect or classify certain objects, along with potential solutions. Lastly, an exploration of our limitations regarding the correlation between distance and reliability is to be expected.

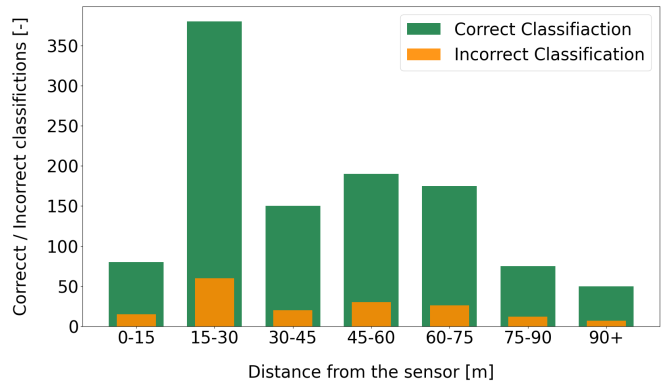


Fig. 4. The number of correct and incorrect classifications on the test dataset plotted against the object’s distance from the sensor. As illustrated above, the distance only slightly affects the ratio of correct to incorrect classifications. This behaviour is the result of our collected samples being sparse to begin with, which greatly aided our classifier in generalizing for longer distances during training.

REFERENCES

- [1] W. Khawaja, M. Ezuma, V. Semkin, F. Erden, O. Ozdemir, and I. Guvenc, “A survey on detection, classification, and tracking of aerial threats using radar and communications systems,” 2024.
- [2] A. Coluccia *et al.*, “Drone-vs-bird detection challenge at iciap 2021,” in *Image Analysis and Processing. ICIAP 2022 Workshops*, P. L. Mazzeo, E. Frontoni, S. Sclaroff, and C. Distanto, Eds. Cham: Springer International Publishing, 2022, pp. 410–421.
- [3] A. Coluccia, M. Ghenescu, T. Piatrik, G. De Cubber, A. Schumann, L. Sommer, J. Klatte, T. Schuchert, J. Beyerer, M. Farhadi, R. Amandi, C. Aker, S. Kalkan, M. Saqib, N. Sharma, S. Daud, K. Makkah, and M. Blumenstein, “Drone-vs-bird detection challenge at ieee avss2017,” in *2017 14th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*, 2017.
- [4] J. Liu, Q. Xu, and W. Chen, “Classification of bird and drone targets based on motion characteristics and random forest model using surveillance radar data,” *IEEE Access*, 2021.
- [5] S. Gazdag, T. Möller, T. Filep, A. Keszler, and A. L. Majdik, “Detection and tracking of mavs using a lidar with rosette scanning pattern,” 2024. [Online]. Available: <https://arxiv.org/abs/2408.08555>
- [6] A. Hornung, K. Wurm, and M. Bennewitz, “Octomap: an efficient probabilistic 3d mapping framework based on octrees,” *Autonomous Robots*, 2013.
- [7] Z. Liu, P. van Oosterom, J. Balado, A. Swart, and B. Beers, “Data frame aware optimized octomap-based dynamic object detection and removal in mobile laser scanning data,” *Alexandria Engineering Journal*, 2023.
- [8] S. Dogru and L. Marques, “Drone detection using sparse lidar measurements,” *IEEE Robotics and Automation Letters*, 2022.
- [9] G. Jocher, “Yolov5 by ultralytics,” 2020. [Online]. Available: <https://github.com/ultralytics/yolov5>

- [10] I. Catalano, H. Sier, X. Yu, T. Westerlund, and J. P. Queralta, "Uav tracking with solid-state lidars: Dynamic multi-frequency scan integration," in *2023 21st International Conference on Advanced Robotics (ICAR)*, 2023.
- [11] H. Sier, X. Yu, I. Catalano, J. P. Queralta, Z. Zou, and T. Westerlund, "Uav tracking with lidar as a camera sensors in gnss-denied environments," 2023.
- [12] M. Hammer, M. Hebel, M. Laurenzis, and M. Arens, "Lidar-based detection and tracking of small UAVs," in *Emerging Imaging and Sensing Technologies for Security and Defence III; and Unmanned Sensors, Systems, and Countermeasures*, G. S. Buller, R. C. Hollins, R. A. Lamb, and M. Mueller, Eds., International Society for Optics and Photonics. SPIE, 2018.
- [13] M. Hammer, B. Borgmann, M. Hebel, and M. Arens, "Image-based classification of small flying objects detected in LiDAR point clouds," in *Laser Radar Technology and Applications XXV*. International Society for Optics and Photonics, 2020.
- [14] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, "Pointnet: Deep learning on point sets for 3d classification and segmentation," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [15] M. Jaderberg, K. Simonyan, A. Zisserman, and k. kavukcuoglu, "Spatial transformer networks," in *Advances in Neural Information Processing Systems*, 2015.
- [16] B. Zhou, H. Zhao, X. Puig, S. Fidler, A. Barriuso, and A. Torralba, "Scene parsing through ade20k dataset," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017.
- [17] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. u. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017.
- [18] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, and O. Beijbom, "Pointpillars: Fast encoders for object detection from point clouds," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.